

RArpad Attila Voros

Curriculum Vitae April 29th, 2026

CONTACT INFORMATION

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Github: <https://github.com/arpadav>

Gitlab: <https://gitlab.com/arpadav>

SUMMARY STATEMENT

Electrical and Computer Engineer with deep experience in signal processing, statistical analysis, and machine learning applied to real-world defense systems. Built performance-critical embedded and real-time platforms in Rust and C++, later extending into full-stack architecture and SaaS product development.

Currently seeking opportunities to operate in fast-paced, high-impact environments, whether through founding, early-stage engineering roles, or technically driven ventures, where end-to-end ownership, rapid iteration, and bridging advanced algorithms with real-world deployment are critical.

PROFESSIONAL EXPERIENCE

Stealth Startup

Charlotte, North Carolina / Salt Lake City, Utah

Founder

March 2026 – Present

- Architecting a Rust-based platform that ingests multi-modal edge video/audio and transforms it into structured, schema-validated reports using vision-language models for scene understanding and ASR for transcript extraction.
- Designing a real-time cross-modal processing pipeline implementing adaptive frame sampling, timestamp-aligned audio/video fusion, domain-constrained template mapping, and deterministic PDF generation for inspection and public-sector reporting workflows.

Udon

Charlotte, North Carolina

Co-Founder

November 2025 – March 2026

- Full-stack restaurant reservation platform to limit and prevent reservation scalping at high-demand fine-dining restaurants. Written in Rust (Dioxus), CockroachDb, and more.

The Johns Hopkins Applied Physics Laboratory

Laurel, Maryland

Advisor(s) — Dr. David Jansing, Dr. Christopher Gifford, Dr. Myron Z. Brown

April 2022 – November 2025

Research Engineer / Team Lead — Imaging Systems / Asymmetric Operations

Overview

- **Algorithms** — 1 / 2 / 3 / N-D signal processing + filtering, time-series analysis, computer vision (classification, detection), deformable registration, multi / hyperspectral re-identification, remote sensing, 2 / 3-D segmentation, path planning, task decomposition, pixel + world-space tracking, photogrammetric 3-D reconstruction, clustering algorithms
- **Data** — imagery (EO / IR / multispectral / hyperspectral / synthetic aperture radar), FMV (EO / IR / multispectral), satellite imagery (various modalities), radar, synthetic data generation (various modalities), pseudo-random walks
- **Engineering Practices** — hardware acceleration, algorithm optimization, benchmarking, low-level programming, memory-safe programming, unit testing, documentation, CI/CD, Dev Ops, ML Ops, fine-tuning, hyper-parameter sweeps, metrics analysis, trade-off analysis, low-latency / high-bandwidth communications, shared-resource constraint considerations, low size-weight-power constraint considerations, portability / scalability / reusability, software hardening, standardization, open-source contribution

Artificial Intelligence & Machine Learning

- Designed and oversaw end-to-end development from conception to deployment, which includes data curation, model design / selection, model training / finetuning, model pruning / optimization, and accounting for resource constraints (size, power draw, memory usage) on embedded devices using C++, Rust, and CUDA for latency-sensitive applications.
- Developed custom deep-learning architectures (PyTorch, JAX, TensorRT) for super-spectral and super-spatial resolution satellite imagery, natural disaster damage assessment of EO / multispectral imagery, Fourier-neural operators for synthetic aperture radar image-formation, and geo-tempo-spatial data analysis of hyperspectral imagery.
- Applied state-of-the-art deep-learning models for detection & classification of data of various modalities (synthetic aperture radar imagery, EO / IR, signals), neural-radiance fields on sparse areas-of-interest, feature-extraction to aid in geo-spatial tracking, and deformable registration of satellite imagery.
- Worked with various USG / DoD sponsors (Navy, Air Force, NASA, IARPA, NGA, other IC organizations)

Technical Development, Integration, Testing, & Deployment

- Developed performance-critical Rust / C++ systems for real-time sensor data on embedded platforms (ARM / x86), optimizing latency / memory.
- Development lead and Rust SME on a modular-in-development, monolithic-in-execution autonomous system. Oversaw ~20 developers and advised government sponsors and industry partners.
- Refactored legacy code into memory-safe Rust / C++ using custom FFI for zero-overhead bindings.
- Architected GPU-accelerated algorithms (CUDA / cuBLAS) for processing, noise reduction, and pattern recognition.
- Spearheaded promotion of software practices and automation (via CICD) of unit-testing, doc-testing, run-time scenario testing, thorough documentation, report generation, and automatic building with Docker / Podman containerization.

North Carolina State University

Advisor(s) — Dr. Tianfu (Matt) Wu

Raleigh, North Carolina

August 2021 – December 2021

Independent Study — Zero-Shot Learning

- Proposed a novel, dynamic deep-learning architecture for few-shot learning classification tasks using feature vectors.
- Analyzed and implemented ZSL dependency of said architecture with semantically meaningful latent space autoencoder.

North Carolina State University ECE Department

Advisor(s) — Dr. Rachana Gupta, Jeremy Edmonson

Raleigh, North Carolina

August 2021 – December 2021

Senior Design Lab TA — Troxler Design Center

- Aided, informed, and serviced students with their senior capstone design project as well as related electronic equipment, components, and tools. Worked in junction with NCSU's ECE Department for lab recommendations & renovations
- Responsible for tens of thousands of dollars' worth of laboratory equipment and upkeep of NCSU Troxler Design Center

North Carolina State University, U.S. Army Research Office

Advisor(s) — Dr. Skip Scheifele, Dr. Rachana Gupta, Dr. Shephard Pitts, Paul Reid

Raleigh, North Carolina

August 2020 – May 2021

Team Lead — Senior Capstone Project (VADER)

- Led a team of 5 in production of a directional acoustic device for deterring African elephants from farmland.
- Designed, simulated, and prototyped multiple device solutions for the U.S. Army Research Office.
- Optimized modulation techniques in minimizing harmonic distortion, developed predistortion-distortion spectrum mapping to further improve quality of sound, simulated various directional-sound propagation techniques in MATLAB.
- Simulated analog load characteristics & hysteresis in LTSpice, designed PCBs in KiCad

North Carolina State University

Advisor(s) — None

Raleigh, North Carolina

September 2018 – May 2021

Treasurer & Committee Member — NCSU PackHacks

- Led and helped organize the 2nd largest free hackathon in the state of North Carolina — <https://ncsupackhacks.org/>
- Developed budgets, acquired annual sponsorship, and managed & distributed all funds for the PackHacks event

Hochschule Reutlingen

Advisor(s) — Dr. Bernd Thomas

Reutlingen, Germany

January 2020 – March 2020

Undergraduate Researcher — Hybrid Energy Modeling

- Optimized Simulink and MATLAB simulations of a hybrid energy system, consisting of energy storage devices (batteries & TES) and energy transfer units (PVs & heat pumps), according to the Klucher weather model
- Ensured Simulink and MATLAB simulations were identical by finding mistakes of both models

Duke Energy Carolinas

Advisor(s) — Glen Frix, Tracy Blackmon

Charlotte, North Carolina

May 2019 – August 2019

Summer Intern — Transmission Engineering

- Created tool using VBA in MS Access which autogenerates SQL queries to find delta in external modeling data, consisting of 5 of the major neighboring energy distributors with thousands of line-connections each.
- Used said VBA tool to automate update of Duke's modeling system.
- Wrote a script in Perl which generated over 100 clean one-line displays for unmodeled 230kV-500kV lines.

Bravo Team LLC

Advisor(s) — Dr. Joshua Tarbutton

Charlotte, North Carolina

December 2018 – January 2019

Winter Break Intern — Engineering Consulting

- Worked on translating VB shot peening simulation for aerospace product manufacturer to Qt to be furthered in development on mobile platforms.
- Selected precision parts for pick-and-place SCARA robot, commissioned by same aerospace product manufacturer
- Constructed CAD models multiple variations of said SCARA robot in SolidWorks

North Carolina State University, Duke University

Raleigh, North Carolina

Advisor(s) — Dr. Robert Golub, Dr. Vince Cianciolo

September 2017 – August 2018

Undergraduate Researcher — nEDM Sensing Apparatus

- Worked on the nEDM intercollegiate experiment for the DOE, working at NCSU and Duke under ORNL.
- Utilized multi-axis translational stage to displace position of a wavelength shifting fiber relative to SiPM to calculate precision installation requirement of “fiber-SiPM” coupling. Maximum tolerance of mounting to be used in Monte-Carlo simulation to estimate rigidity specifications of sensor containment unit used in the nEDM experiment at ORNL.

University of North Carolina at Charlotte

Charlotte, North Carolina

Advisor(s) — Dr. Joshua Tarbuton

May 2017 – August 2017

Team Member — Voluntary Summer Research

- Designed a desktop CNC milling machine for high-speed machining.
- Utilized polar coordinates as opposed to Cartesian in machine design. Precision rotary table was used to reduce bed size.
- Created CAD models, conducted stress tests using Autodesk Inventor, and partook in thousand-dollar decision-making.

Intel International Science and Engineering Fair

Waxhaw, North Carolina

Advisor(s) — None; but special thanks to Dr. Faramarz Farahi

November 2016 – May 2017

Team Lead & Independent Researcher — Muon Scattering Tomography

- Led an independent research team of 3 to reduce the cost of conventional muon scattering tomography by 96%.
- Acquired provisional patent for novel approach, which utilizes volumetric scintillators and a trilateration algorithm.
- Built a semi-functional prototype. Sensing provided by SiPM arrays coupled with scintillating. Created Monte-Carlo and signal-processing simulations using Java, MATLAB, and LTSpice.
- Responsible for thousands of dollars’ worth of equipment. No external funding was provided.

PROJECTS

For a full list of personal, academic, and professional projects with descriptions, figures, and interactivity, please see:

<https://arpadvoros.com/projects/>

EDUCATION

North Carolina State University

Raleigh, North Carolina

Master of Science Electrical Engineering

2020 – 2021

GPA: 3.96/4.00

North Carolina State University

Raleigh, North Carolina

Bachelor of Science Electrical Engineering Summa Cum Laude

2017 – 2020

Major GPA: 3.97/4.00 Cum. GPA: 3.76/4.00

USG CLEARANCE(S)

TS / SCI

June 2023 – Present

Secret

November 2022 – Present

SUMMARY OF SKILLS

- **Programming Languages** — Rust, Python, C++, C, MATLAB, Haskell, Java, JavaScript, Perl, Verilog, R, SQL
- **Software Tooling / Skills** — Linux, ONNX, TensorRT, Python Libraries (PyTorch, TensorFlow, scikit-learn, pandas), Apache Airflow, LaTeX, KiCad, LTSpice, PSpice, 2-3D CAD
- **Interests** — signal processing, quantitative research, financial markets, image processing, computer vision, deep learning, applied machine learning, information theory, data analytics & visualization, embedded systems, analog circuit design, radio, optics, acoustics, human-computer interaction, brain-computer interface
- **Relevant graduate courses** — ECE 763: Computer Vision, ECE 558: Digital Imaging Systems, ECE 633: Individual Topics in ECE, ECE 542: Neural Networks, ECE 513: Digital Signal Processing, ECE 514: Random Processes, ECE 560: Embedded Systems Architecture, ECE 592: Introduction to Satellites, MA 405: Linear Algebra, ECE 498: Special Projects, ECE 574: Computer and Network Security, ECE 564: ASIC & FPGA Design
- **Hands-On Skills** — rapid prototyping, extensive electronics and physics laboratory experience, machining
- **Languages** — English (fluent), Hungarian (fluent), German (conversational proficiency)

PRESENTATIONS

- Voros, Arpad. (2021, December). *Analysis and Implementation of a Semantic Auto-Encoder for Zero-Shot Learning*. North Carolina State University. Raleigh, North Carolina
- Voros, Arpad., Cook, Hunter., Alamro, Nwaf., Fitts, Greyson. Pyrtle, Morgan. (2020, November). *Senior Design Day Team 21 – Vectorized Acoustic Deterrence of Elephants Research*. North Carolina State University. Raleigh, North Carolina

- Voros, Arpad., Daino, Trevor., Kronovet, Michael. (2017, May). *PHYS024T – Muon Scattering Tomography: Utilizing Silicon Photomultiplier Arrays to Trilaterate Muon Multiple Coulomb Scattering Events*. Intel International Science and Engineering Fair. Los Angeles, California.

AWARDS & HONORS

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| <i>Coin Award (x4), REDD Bravo Award</i> | <i>The Johns Hopkins U. Applied Physics Lab. – 2022 – 2025</i> |
| ▪ Misc. awards internal to JHU APL for leadership, initiative, and technical communication. | |
| <i>Semester Dean's List (x7)</i> | <i>North Carolina State University – 2017 – 2021</i> |
| ▪ Earning a semester GPA of 3.5 or greater on 12 – 14 credit hours of coursework, or 3.25 or greater on 16 or more credits | |
| <i>1st, ECE Senior Design Day</i> | <i>North Carolina State University – April 2021</i> |
| ▪ Senior design team received first place in NCSUs ECE Senior Design Day competition for outstanding project, prototype demonstration, and presentation. | |
| <i>Perfect Pitch Award 1st Place Winner</i> | <i>North Carolina State University – November 2020</i> |
| ▪ Senior design team received first place of over 140 students in having the best poster and best three-minute pitch in describing their project. | |
| <i>ASPE 32nd Conference NSF Grantee</i> | <i>ASPE – September 2017</i> |
| ▪ Received grant from National Science Foundation to cover attendance costs for the 32nd Annual ASPE (American Society for Precision Engineering) Conference at Charlotte, NC in November 2017. | |
| <i>Third Award, Physics and Astronomy, Intel ISEF</i> | <i>Society for Science & the Public – May 2017</i> |
| ▪ Third Award at Intel ISEF for \$1,000 in the Physics and Astronomy category. | |
| <i>Intel Excellence in Computer Science Award</i> | <i>Intel Foundation – February 2017</i> |
| ▪ Received recognition and a prize of \$200 for the original development of a Monte Carlo simulation in the Java and MATLAB languages to model the efficacy of a novel approach to conducting muon scattering tomography. The simulation modeled the propagation of muons, their angular distribution, through scintillating prisms, and through high-Z material cross-sections, real-time electronic signal read-out of SiPM, and thermal noise characteristics of SiPM. | |
| <i>1st, UNC Charlotte Excellence in Physics</i> | <i>Physics Department at UNC Charlotte – February 2017</i> |
| ▪ Received 1st place distinction and a prize of \$100 on behalf of the demonstration of sound physics concepts in the design, construction, calibration, and simulation of a novel technique for conducting muon scattering tomography. | |
| <i>Region 6 NCSEF 2017 1st Place Winner, ISEF Finalist</i> | <i>The Center for STEM Education – February 2017</i> |
| ▪ Nominated and named finalist for the Intel International Science and Engineering Fair 2017 in Los Angeles, California. | |

PROFESSIONAL ASSOCIATIONS

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| IEEE (Institute of Electrical and Electronics Engineers) | <i>December 2024 – Present</i> |
| SPiE (Society of Photo-Optical Instrumentation Engineers) | <i>March 2024 – Present</i> |
| ASPE (American Society for Precision Engineering) | <i>October 2017 – October 2018</i> |
| Science National Honors Society | <i>September 2014 – June 2017</i> |
| Mu Alpha Theta | <i>August 2014 – June 2017</i> |
| German Honors Society | <i>August 2013 – June 2017</i> |

REFERENCES

Available upon request